

Anomaly Network Monitoring for Software-Defined Networking

Shengping Wang*

Electronic Information School, Wuhan University

Wuhan, China

*2474036202@qq.com

Abstract—With the rapid iteration of information technology, the network field is undergoing an unprecedented transformation wave. Especially in the current era of surging dynamic traffic, the limitations of traditional network architectures have become increasingly prominent. Software-Defined Networking (SDN), a disruptive and innovative network paradigm, addresses the tight coupling of control and forwarding in traditional networks but the centralized design carries potential security risks: a compromised core controller (via malicious intrusion or traffic flooding) paralyzes the entire network's decision-making and scheduling, causing cascading service disruptions. To tackle anomaly traffic monitoring, this paper proposes an SDN defense mechanism based on Origin-Destination (OD) flow record analysis, adopting a Deep Neural Network (DNN) for anomaly detection to enable accurate responses and reduce attack impacts. Experiments on the public CICDDoS2019 dataset compared DNN with Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM), evaluating each via performance tests. The experimental results demonstrate DNN achieves an excellent detection rate, accurately identifying anomaly traffic and proving a feasible solution for the proposed system.

Keywords—Software-Defined Networking (SDN); Anomaly Traffic Detection; Deep Neural Network (DNN)

I. INTRODUCTION

As information technology evolves rapidly, the network sector is witnessing an unprecedented transformation—among which the sharp surge in dynamic traffic stands out as one of the most distinctive features. The fluctuation of this dynamic traffic is not a simple linear growth, but presents explosive, sudden, and fragmented patterns. Severe traffic fluctuations pose great challenges to network bandwidth, data transmission capabilities, etc. Traditional static network architectures can hardly meet the demand. Therefore, Software-Defined Networking (SDN) [1], with its core features such as "separation of control and forwarding", "centralized management", and "programmability", has proven to offer significant advantages, yet cyberspace security issues remain the core challenge. This layered architectural feature makes the security threats [2] it faces more complex. As an important carrier of security threats, anomaly traffic is often accompanied by abnormal traffic characteristics. Particularly in the SDN architecture, due to the dynamics of traffic scheduling and the flexibility of network topology, anomaly traffic is harder to detect and spreads faster. If not detected in a timely manner, it may spread to the entire network in a short period. Therefore, abnormal traffic detection has become a key component in ensuring network security.

Traditional cybersecurity solutions are limited by slow response, low protection efficiency and high false alarm rates. In contrast, deep learning, a machine-learning approach based on deep neural network models, has brought innovative solution [3] to cybersecurity defense. It can mine hidden patterns from massive information to cope with the evolution of new threats, and has high processing efficiency. With the acceleration of CPU hardware, deep learning enables real-time analysis of high-dimensional traffic data in real time to meet the detection needs of dynamic scenarios [4].

This paper proposes an anomaly traffic monitoring and defense system for SDN based on OD flows. The system is mainly divided into two modules: detection and defense.

The detection module aims to identify anomaly traffic. This module employs the Deep Neural Network (DNN) algorithm as its classifier. Through multi-dimensional feature extraction, the algorithm analyzes individual OD flows and operates as a binary classifier to classify OD flows into normal or abnormal categories.

The defense module generates effective countermeasures against detected attacks. The detection mechanism proposed in this paper is based on OD flows, which can identify abnormal OD flows in real time, swiftly locate the source of attacks, and applies tiered handling to the identified anomalies. Therefore, we introduce a targeted mitigation approach aimed at rapidly detecting and defending against malicious traffic.

To evaluate the efficiency of DNN (Deep Neural Network) as a detection method, we compared DNN with the comparative algorithms CNN and LSTM by using the publicly CICDDoS2019 dataset [5], focusing primarily on how well these deep learning approaches handle various Distributed Denial-of-Service (DDoS) attack scenarios.

The main contributions of this paper can be summarized as follows:

- Proposed a new SDN defense mechanism;
- Employed DNN deep learning;
- Conduct comparative tests with different algorithms on a public dataset to verify the feasibility of DNN in monitoring anomaly traffic.

The structure of this paper is arranged as follows: Chapter 2 describes the related work on SDN networks and anomaly traffic identification; Chapter 3 introduces SDN networks, especially their new three-layer architecture; Chapter 4 outlines

the algorithm flow and model used; Chapter 5 presents the specific simulation experiments; Chapter 6 concludes with a summary.

II. RELATED WORK

SDN is a new network paradigm that subverts traditional network architectures. With the separation of the control plane and the data plane, it enables managers to achieve flexible network management through centralized control logic, and has absolute advantages especially in dynamic traffic allocation. SDN technology has found applications across various fields offering corresponding solutions, such as the application in the internet of Things (IoT) [6][7]. Researchers have pointed out the advantage of SDN in resource scheduling [8].

While centralized management and control is a core advantage of SDN, it can lead to significant losses if subjected to attacks or operational errors. Therefore, the security of the controller has become critical in SDN deployment. There have been many defense systems [9]. However, these defense systems still have shortcomings in identification accuracy and anomaly traceability, and they are likely to disrupt normal traffic.

In a network environment, anomaly traffic is generally defined as data flow that deviates from the regular behavior pattern, and its core feature lies in the significant difference from normal network activities. Such traffic is often a major vector of security risks, and can be specifically divided into malicious attack traffic and non-malicious anomaly traffic. For SDN networks, due to the flexibility of their architecture and centralized control characteristics, real-time monitoring and identification of malicious traffic have become a key defense method to ensure network security. Currently, several methods are already available for detecting abnormal traffic (e.g., IP flow-based methods, etc.).

In response to this, this paper proposes a novel SDN defense mechanism, conducts experiments using several main deep learning algorithms—DNN, CNN [10], and LSTM [11]—and ultimately puts forward a monitoring approach based on individual OD flows [12] to quickly detect and mitigate the impact of attacks. The proposed detection method leverages DNN (Deep Neural Network) for analysis to identify attacks on Software-Defined Networking.

III. SDN SECURITY FRAMEWORK

This section elaborates on the working principle of the proposed SDN defense system. Although SDN has the advantage of centralized management, its single control center makes it vulnerable to attacks—any attack or intrusion could potentially lead to network paralysis. Hence, it is vital to ensure the security of the controller to maintain the overall network's safe and reliable operation. To address this challenge, this paper proposes an SDN security defense system, by analyzing individual OD flows, which can effectively detect various attacks on the central controller.

This defense system targets "single-flow-level" anomalies, detects individual OD (origin-destination) flows to achieve precise control. Different from traditional detection methods, it will not interface with or restrict other normal traffic. The

primacy drawbacks of this approach are high computational overhead and poor interpretability. However, compared with conventional methods, this approach retains a significant advantage in rapid detection, effectively reducing the impact on normal users while realizing user identity identification—because OD flows contain both the Origin address and Destination address, which can quickly lock the specific source and target of anomaly traffic.

Our SDN security framework consists of two core submodules: a detection module and a defense module. These two modules interact via the central system logic, and the entire SDN defense system runs within the SDN controller itself. All data analysis is performed automatically enabling network administrators to simply receive notifications of detected attack events.

The detection module can be divided into two sub-modules. The first sub-module is responsible for raw data collection and feature extraction. It receives flow table statistical data reported by switches/routers in the data plane, extracts the basic identification and raw data traffic of individual OD flows, generates input feature vectors containing a total of 59 features (such as flow identification, flow duration and packet statistics, flow rate and time interval, packet identification and header, packet length distribution, TCP flag statistics, etc.), and transmits the feature vectors to the second sub-module. The second sub-module serves as the DNN anomaly detection module, which is responsible for determining whether an individual OD flow exhibits abnormal behavior. It uses a binary classifier to output either 0 or 1, signaling whether the flow is normal or anomalous and transmits the signal to the defense module. If the traffic is determined to be normal, it operates normally; if it is determined to be abnormal, the defense mechanism is triggered. This sub-module is the main module for realizing SDN anomaly traffic detection, and its specific implementation will be explained in the further chapter.

The defense module is responsible for receiving OD flows determined to be anomaly traffic by the detection module, defining and adopting optimal countermeasures to minimize the impact of attacks on SDN. The main measures adopted are directly discarding the anomaly traffic or diverting it to an alternative route. This module is not the focus of this study and will not be elaborated in detail.

It should be emphasized that this SDN defense system targets individual OD flows, takes the communication parties as the core identifier. It groups scattered data packets into an independent unit, essentially representing a "logical integration" of network traffic. Compared with traditional traffic detection based on ports, links, or devices, it is more refined, thus realizing precise fine-grained management and control. In addition, this defense system can achieve automatic response ultimately demonstrating the effectiveness of this SDN defense system.

IV. ALGORITHM DESIGN

The focus of this study is to improve the accuracy of anomaly traffic detection. Therefore, this chapter will specifically explain the deep learning algorithm implemented in this SDN defense system. The core advantage of deep learning

methods lies in their ability to deeply mine and dynamically model the complex features of single-flow-level. As an end-to-end independent traffic unit, an individual OD flow contains nonlinear time-series features such as flow rate fluctuations, packet interval distribution, and byte length variations. These features are often difficult to be fully captured by traditional rules or statistical methods. However, deep learning techniques like DNN, CNN and LSTMs can automatically learn the temporal dependencies of single-flow data and accurately identify hidden anomalies. Moreover, models such as DNN and Transformer can extract high-level associated features from massive single-flow features without manual feature engineering. This adaptability to complex, dynamic, and sparse features of single flows makes deep learning an efficient technical path for anomaly traffic monitoring based on individual OD flows. Especially in the SDN framework, integrating global single-flow data from the control layer further enhances detection accuracy and real-time responsiveness.

Among the various deep learning methods, the deep neural networks (DNNs) show unique advantages in detecting SDN anomaly traffic, primarily reflected in its superior ability in mining high-level associations of static features and extracting local features compared with other methods such as CNN and LSTM. This paper proposes to apply the DNN-based method to detect anomaly traffic in the SDN environment, and specifically adopts a 2-layer fully connected neural network.

The calculation essence of DNN is the layer-by-layer propagation of linear transformation followed by nonlinear activation. In the forward propagation stage, linear transformation (1) is first performed on the input features, and then nonlinearity is introduced via the ReLU function. The use of ReLU as the activation function (2) here helps alleviate the gradient vanishing problem:

$$z_1 = XW_1 + b_1 \quad (1)$$

$$a_1 = \text{ReLU}(z_1) = \max(0, z_1) \quad (2)$$

Then, linear transformation (3) is performed on the output of the hidden layer, and the result is mapped to the range of 0–1 through the Sigmoid function (4) whose output value is between 0 and 1, which can be directly interpreted as the "probability of belonging to the positive class (attack)" to yield the prediction probability:

$$Z_2 = a_1W_2 + b_2 \quad (3)$$

$$a_2 = \text{Sigmoid}(z_2) \quad (4)$$

In this defense system, binary_crossentropy is specified as the loss function (5)(6) to measure the difference between the predicted probability and the true label. The formula is as follows:

$$L = -[y \log(a_2) + (1 - y) \log(1 - a_2)] \quad (5)$$

$$J = \frac{1}{n} \times \sum_{i=1}^n L_i \quad (6)$$

During the backpropagation stage, Adam is used as the optimizer. Adam is an optimization algorithm based on gradient descent, which updates weights and biases through an adaptive learning rate. At its core, it involves computing gradients and iteratively refining the parameters:

Gradient calculation: Calculate the partial derivatives of the loss J with respect to each parameter (W_1, b_1, W_2, b_2) through backpropagation.

Parameter update (7)(8): the parameter η is the adaptive learning rate; Adam dynamically adjusts η through momentum and second-moment estimation to balance convergence speed and stability.

$$W = W - \eta \frac{\partial J}{\partial W} \quad (7)$$

$$b = b - \eta \frac{\partial J}{\partial b} \quad (8)$$

The complete process of the DNN algorithm can be divided into three stages: data preprocessing, model training, and model prediction & evaluation.

V. SIMULATION EXPERIMENTS

This section focuses on analyzing the performance of the detection module in the defense system. We conducted research using the DNN method and verified the performance advantages of the proposed scheme by comparing it with mainstream detection methods such as CNN and LSTM. These methods were selected based on related work as they represent effectiveness and feasibility in application environments similar to that proposed in this work.

All methods were implemented on the PaddlePaddle AI Studio cloud development platform (2 Cores CPU, 8GB RAM) using Python and Keras (for DNN, CNN, LSTM). This project adopts a Python environment with Python 3.7.13, paired with NumPy 1.16.4, Scikit-learn 0.21.2, Pandas 0.24.2, TensorFlow 1.14.0, Keras 2.2.4, Matplotlib 3.1.0, and Seaborn 0.11.2.

The DNN adopts a single hidden layer containing 2 neurons. The CNN adopts a 3-layer Conv1D convolution structure, using 64, 32, and 16 filters respectively, with kernel sizes of 16, 8, and 3. The LSTM consists a single-layer LSTM structure containing 32 neurons. Both CNN and LSTM are followed by a Dropout layer with a dropout rate of 0.5, and the final output layer has 1 neuron, using the sigmoid activation function for binary classification.

To compare the efficiency of the test methods, we used traditional classification metrics such as Accuracy, Precision, Recall, and F-score. Accuracy represents the percentage of correctly classified traffic records. Precision measures the proportion of IP traffic that is correctly identified as abnormal among all samples classified as abnormal. The Recall metric estimates the percentage of correctly identified anomaly traffic. The F-score represents the harmonic mean between the true positive rate (malicious traffic classified as anomalous) and Precision. In addition, we also compared the Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), normal traffic identification rate, and attack traffic identification rate to further assess the error degree

between predicted values and true values and the performance of the three methods.

This performance comparison was mainly conducted on the public CICDDoS2019 dataset. The CICDDoS2019 public dataset was generated by constructing experimental scenarios close to real network environments—first simulating various modern DDoS attacks, then using professional tools to capture network traffic logs, and parsing the original .pcap format logs into structured tabular data including traffic feature fields and attack labels.

It contains approximately 80–100 official original statistical features. We processed the data and extracted 59 core OD flow features as input, which accurately capture the "source-destination interaction pattern" of network communication and provide targeted and low-redundancy key input for attack detection. 67% of the data was used as the training set, and the remaining 33% as the test set.

Table I shows the exact values of the efficiency metrics of each method.

TABLE I. METRICS OF EACH METHOD

Evaluation Metrics	Model		
	DNN	CNN	LSTM
Accuracy	0.99913255497725	0.9903710253267	0.9818874799884
Precision	0.99916055491738	0.9903501076640	0.9994049988305
Recall	0.99979904748507	0.9981713321142	0.9788477382794
F-Score	0.99947969922999	0.9942453387243	0.9890195568838
Average	0.99939296415242	0.9932844509573	0.9872899434955
MSE	0.00086744502274	0.0096289746732	0.0181125200115
MAE	0.00086744502274	0.0096289746732	0.0181125200115
RMSE	0.02945241964153	0.0981273390716	0.1345827626834

Fig. 1 presents the comparison results of traditional metrics obtained by each method. Herein, we analyzed the accuracy, precision, recall, and F-score of each method; the analysis of variance (ANOVA) and the recognition rate of different types of traffic will be discussed subsequently.

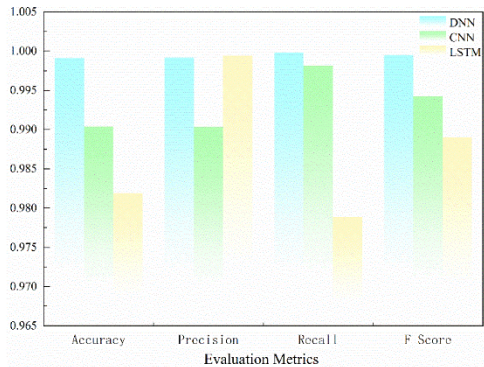


Fig. 1. Individual Results of Accuracy, Precision, Recall, and F-score for Each Method

It can be observed that all three test methods achieved excellent performance, with scores of all metrics being close to full marks exceeding 95%. However, the figure intuitively shows that the Deep Neural Network (DNN) outperforms the Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) across all metrics. In terms of accuracy, DNN is close to 1, reaching 99.91%, while CNN (99.04%) and LSTM (98.19%) are slightly lower. Regarding precision, LSTM is comparable to DNN, reaching 99.9%, whereas CNN (99.04%) performs slightly worse. For recall and F-score, DNN still demonstrates the best performance, achieving 99.98% and 99.94% respectively, followed by CNN (99.81% and 99.42%) and LSTM (97.88% and 98.90%). Since DNN outperforms CNN and LSTM in all metrics, there is no doubt that DNN is overall superior in terms of traditional metrics.

Next, we compared the secondary metrics, namely MAE, MSE, and RMAE. Fig. 2 below lists the specific results, from which it can also be clearly observed that the MSE and MAE of DNN are significantly lower than those of CNN and LSTM. In terms of RMSE, DNN falls in the middle. On average, the Deep Neural Network (DNN) still achieves the best performance with the lowest model error rate.

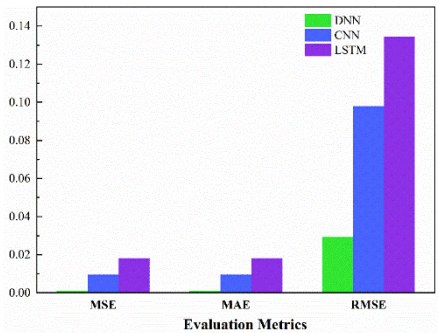


Fig. 2. MSE, MAE, and RMSE of Various Methods

Finally, we also compared the correct recognition rates of the three methods for normal traffic and abnormal traffic, and the results are shown in Fig. 3.

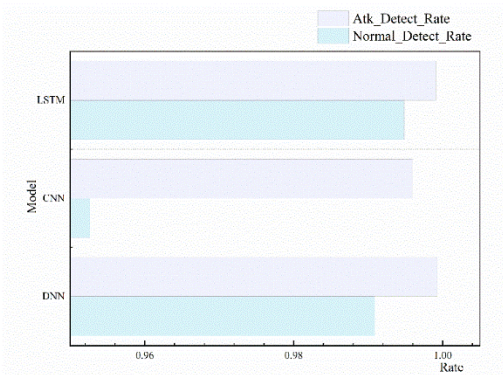


Fig. 3. Correct Recognition Rates for Different Types of Traffic

We can observe that the correct recognition rates of DNN for both normal and abnormal traffic have exceeded 99.5%. However, CNN and LSTM exhibit significantly poor performance in the correct recognition of normal traffic and abnormal traffic respectively, with rates lower than 98%.

Through the above evaluation, all three methods have demonstrated high accuracy in recognizing individual OD flows, confirming their feasibility in detecting abnormal network traffic in SDN. Notably, DNN delivered the best recognition effect and outperforms the other compared algorithms across all metrics. Additionally, it is worth noting that the DNN used in this method only has a single hidden layer (with 2 neurons), thus possessing a significant computational advantage over CNN and LSTM.

Therefore, through this simulation experiment, selecting DNN for the detection module in our proposed SDN defense mechanism is deemed the optimal approach.

VI. CONCLUSION

This paper proposes a defense mechanism against abnormal traffic in Software-Defined Networking (SDN), which can accurately identify abnormal traffic, effectively detect controller-targeted network attacks and mitigate critical network threats. Based on single OD flow detection, the mechanism consists of two core modules: detection and defense. The detection module, the core component, is split into two sub-modules for OD flow extraction and binary traffic classification (normal/abnormal); the defense module receives defense signals and processes abnormal traffic accordingly. This mechanism can effectively reduce the impact of network attacks and ensure the stable operation of SDN.

Simulation experiments on the public CICDDoS2019 dataset compare the Deep Neural Network (DNN) with CNN and LSTM for abnormal traffic detection. The results confirm that all three models are applicable for detection tasks, while DNN achieves superior performance in all metrics with higher accuracy and precision.

REFERENCE

- [1] McKeown N, Anderson T, Balakrishnan H, et al. OpenFlow: enabling innovation in campus networks [J]. *ACM SIGCOMM Computer Communication Review*, 2008, 38 (2): 69-74.
- [2] Rahouti M, Xiong K, Xin Y, et al. SDN security review: Threat taxonomy, implications, and open challenges[J]. *Ieee Access*, 2022, 10: 45820-45854.
- [3] Fishwick P A. Neural network models in simulation: A comparison with traditional modeling approaches[C]//*Proceedings of the 21st conference on Winter simulation*. 1989: 702-709.
- [4] Naseer S, Saleem Y, Khalid S, et al. Enhanced network anomaly detection based on deep neural networks[J]. *IEEE access*, 2018, 6: 48231-48246.
- [5] Sharafaldin I, Lashkari A H, Hakak S, et al. CSE-CIC-IDS2018: A comprehensive dataset for network traffic intrusion detection and malicious activity recognition [J]. *arXiv preprint arXiv:1807.04467*, 2018.
- [6] Rabet I, Selvaraju S P, Fotouhi H, et al. SDMob: SDN-based mobility management for IoT networks[J]. *Journal of Sensor and Actuator Networks*, 2022, 11(1): 8.
- [7] Desai P R, Mini S, Tosh D K. Edge-based optimal routing in SDN-enabled industrial Internet of Things[J]. *IEEE Internet of Things Journal*, 2022, 9(19): 18898-18907.
- [8] Rahman A, Islam M J, Montieri A, et al. Smartblock-sdn: An optimized blockchain-sdn framework for resource management in iot[J]. *IEEE Access*, 2021, 9: 28361-28376.
- [9] Alshamrani A, Chowdhary A, Pisharody S, et al. A defense system for defeating DDoS attacks in SDN based networks[C]//*Proceedings of the 15th ACM international symposium on mobility management and wireless access*. 2017: 83-92.
- [10] Xue H, Jing B. SDN attack identification model based on CNN algorithm[J]. *IEEE Access*, 2023, 11: 87652-87666.
- [11] Tayfour O E, Mubarakali A, Tayfour A E, et al. Adapting deep learning-LSTM method using optimized dataset in SDN controller for secure IoT[J]. *Soft Computing*, 2023: 1-9.
- [12] Liu G, Guo S, Xiao B, et al. SDN-based traffic matrix estimation in data center networks through large size flow identification[J]. *IEEE Transactions on Cloud Computing*, 2019, 10(1): 675-690.